

Predicting Kyphosis Disease Using XgBoost Compares Over Random Forest With Improved Accuracy.

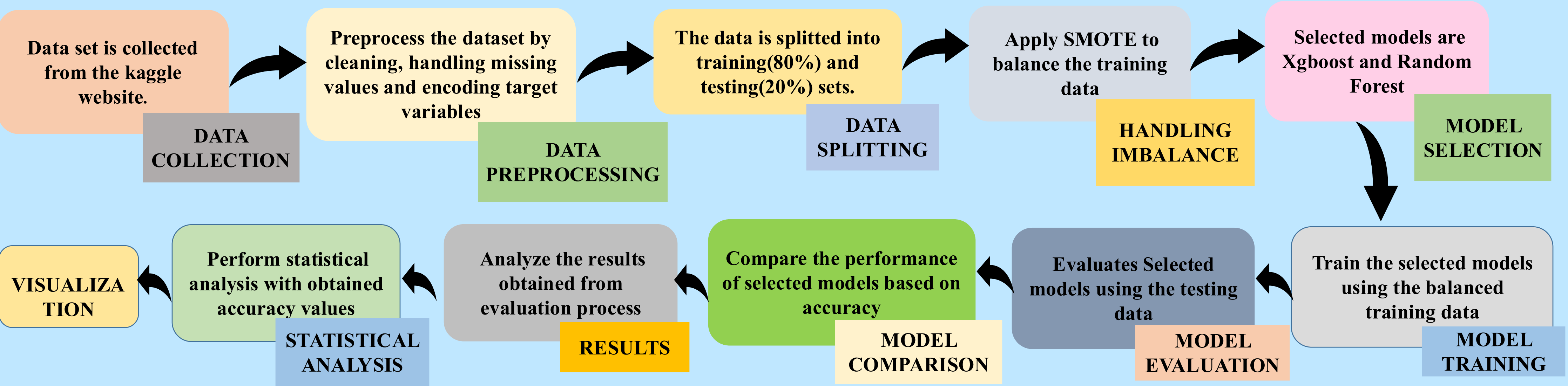
INTRODUCTION

- The aim of this study is to compare the accuracy of predicting kyphosis disease using the XGBoost and Random Forest algorithms.
- It is important to evaluate and compare these algorithms to determine which one yields improved accuracy in predicting kyphosis. This can assist healthcare professionals in making more accurate diagnoses and treatment decisions for patients with kyphosis.
- he findings of this study can be applied in the field of orthopedics and spine care. Accurate prediction of kyphosis can help in early detection, intervention, and personalized treatment plans for patients.
- XGBoost is a powerful gradient boosting algorithm, while Random Forest is an ensemble learning method that uses multiple decision trees.
- The dataset used in this study includes variables such as age, gender, start, and kyphosis. These variables are used to train and evaluate the performance of both the XGBoost and Random Forest algorithms in predicting kyphosis disease.



Fig. 1:Normal and affected spine by kyphosis

MATERIALS AND METHODS



RESULTS

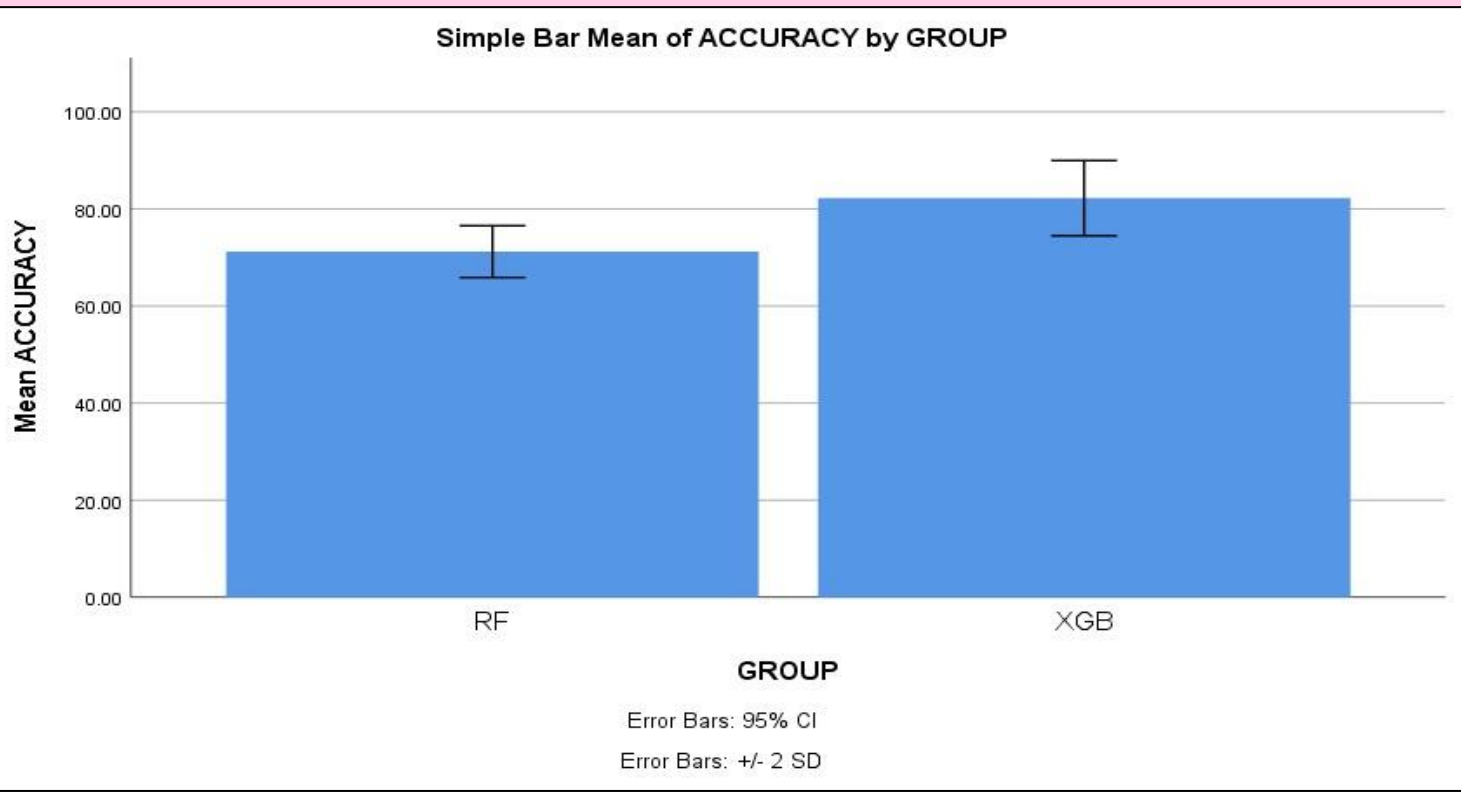


Fig. 2:Comparison of kyphosis detection between Xgboost and Random forest.

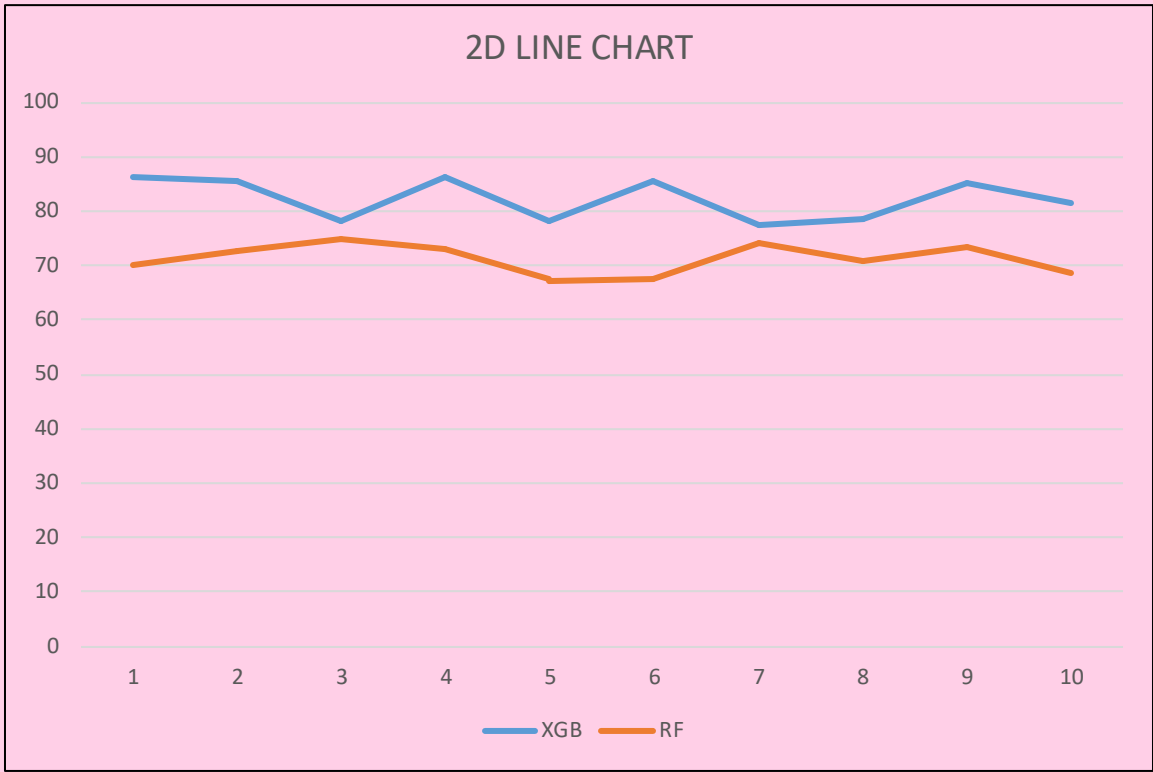


Fig. 3: Comparison of accuracy values of kyphosis detection with Xgboost and Random forest.

GROUP		N	Mean	Std. Deviation	Std. Error Mean
ACCURACY	XGB	10	82.2140	3.87330	1.22484
	RF	10	71.1930	2.67502	0.84591

Table. 1: Comparison of the mean accuracy of kyphosis detection with Xgboost and Random forest.

DISCUSSION AND CONCLUSION

- Based on T-test statistical analysis, the significance value of $p=0.001$ (independent sample T-test $p<0.05$) is obtained and shows that there is a statistical significant difference between the group 1 and group 2 (group 1-Xgboost & group 2-Random forest).
- The Random forest algorithm exhibited an accuracy rate of 70.59%, whereas the XgBoost algorithm, demonstrated an accuracy of 82.35% in predicting kyphosis disease in individuals.
- Expanding the analysis to include ensemble methods and exploring the potential of transfer learning for kyphosis detection in resource-limited settings.
- Age, gender, and other factors can affect the accuracy of kyphosis prediction using machine learning. However, limitations arise from data availability, quality, potential bias, and the need for validation on diverse populations.
- In conclusion, based on the comparison between XGBoost and Random Forest for predicting kyphosis disease, XGBoost demonstrates improved accuracy. This indicates that XGBoost is a promising algorithm that can potentially enhance the accuracy of kyphosis disease prediction.

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